

FEATURE IMPORTANCE REPORT

Customer Churn Prediction – Week 3 Machine Learning Project

1. Introduction

Feature importance was used to identify which input variables contributed most strongly to the predictions made by the Random Forest and XGBoost models. This analysis helps translate the machine learning results into understandable business signals and supports interpretation of the churn-prediction models.

2. Interpretation of Feature Importance

Feature importance represents the relative contribution of a feature to a model's predictive decisions. A higher importance value means that the feature was more useful to the model when making predictions. These values should be interpreted as predictive associations rather than evidence that a feature directly causes customer churn.

3. Random Forest Feature Importance

The Random Forest model identified TotalCharges, tenure, and MonthlyCharges as its three strongest predictors. Contract type, internet service, payment method, TotalServices, and PaperlessBilling were also among the leading features.

Rank	Feature	Importance
1	TotalCharges	0.174278
2	tenure	0.157923
3	MonthlyCharges	0.145814
4	Contract_Two year	0.061937
5	InternetService_Fiber optic	0.044022
6	PaymentMethod_Electronic check	0.037899
7	TotalServices	0.037482
8	Contract_One year	0.027158
9	gender_Male	0.025196
10	PaperlessBilling_Yes	0.023981

4. XGBoost Feature Importance

XGBoost placed particularly high importance on contract type and internet-service variables. Contract_Two year was the most important feature, followed by Contract_One year and InternetService_Fiber optic. PaymentMethod_Electronic check and tenure were also among the leading predictors.

Rank	Feature	Importance
1	Contract_Two year	0.363767
2	Contract_One year	0.178733
3	InternetService_Fiber optic	0.119743
4	InternetService_No	0.075932
5	PaymentMethod_Electronic check	0.040388
6	StreamingMovies_Yes	0.036806
7	tenure	0.036159
8	OnlineSecurity_Yes	0.026336
9	PaperlessBilling_Yes	0.018009
10	StreamingTV_Yes	0.011147

5. Comparison of the Two Models

The two models identify several common predictors, but their rankings differ. Both models place contract type, internet service, payment method, and tenure among important predictive variables. Random Forest gives greater importance to TotalCharges and MonthlyCharges, whereas XGBoost places much stronger emphasis on contract type and internet-service categories.

The strongest common signals are Contract_Two year, Contract_One year, InternetService_Fiber optic, PaymentMethod_Electronic check, tenure, and PaperlessBilling_Yes. Their presence across the models indicates useful predictive information in this dataset, although the models quantify their contributions differently.

6. Key Business Insights

Contract type: Contract variables were particularly prominent in XGBoost. This suggests that customer contract status is an important signal for identifying churn risk and should be examined in retention analysis.

Internet service: InternetService_Fiber optic and InternetService_No were among XGBoost's strongest predictors. These segments should be investigated for differences in service experience, pricing, support, and customer expectations.

Customer tenure: Tenure was among the top predictors in both models. Customer relationship duration can therefore be incorporated into risk segmentation and retention planning.

Charges: TotalCharges and MonthlyCharges were among Random Forest's strongest predictors. Charge-related patterns should be examined when assessing customers' churn risk.

Payment method: PaymentMethod_Electronic check ranked highly in both tree-based models. The business should investigate whether billing or payment experience contributes to elevated churn risk in this segment.

Customer services: OnlineSecurity_Yes, StreamingTV_Yes, StreamingMovies_Yes, and TotalServices appeared among important predictors. These variables can help the business investigate how service bundles relate to customer retention.

7. Business Recommendations

- Use the selected XGBoost model to identify customers with elevated churn risk and prioritize retention actions.
- Investigate contract structure and develop suitable strategies for encouraging longer-term customer relationships.
- Review customer experience, pricing, and support associated with important internet-service segments.
- Investigate electronic-check customers for billing, payment, or service-experience patterns associated with churn.
- Combine tenure and charge information with predicted churn risk when designing customer segments.
- Use feature importance as a guide for further business investigation, not as proof of causal relationships.

8. Limitations

Feature importance is model-specific and does not establish causality. A high importance score indicates that a feature contributed to prediction, not that changing the feature will necessarily change churn. Feature importance can also be affected by relationships among correlated variables. Further analysis would be required to determine causal drivers of churn.

9. Conclusion

The feature-importance analysis provides useful evidence about the variables that the Random Forest and XGBoost models relied on when predicting customer churn. Contract type, internet service, tenure, payment method, and charge-related variables emerged as important predictive signals. XGBoost placed particularly strong emphasis on contract and internet-service features, while Random Forest placed greater emphasis on charges and tenure. These findings can guide targeted retention analysis while recognizing that predictive importance is not equivalent to causation.